

Ekagra Luthra

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Languages & Frameworks

Golang, Typescript (React), Java, Python, Apache Thrift, Kafka, Dart, Linux, C++ , SQL, Git, gRPC

Experience

Workiva | Software Engineer

May 2023 – Current

- Designed and implemented a full **SSE-S3** re-encryption lifecycle for Bring Your Own Key (**BYOK**) metadata by rerouting metadata traffic off an S3 encryption proxy, directly onto **AWS SSE-S3**. Built **Go** object storage short-circuit path, **Java** rekey framework with **SQS** task retries, and rolled out migrations across all production regions. Delivered a 12x per-request cost reduction.
- Led a migration from **New Relic** to **Datadog** for observability for a 3-service BYOK encryption platform stack, shipping parity dashboards and **SLOs** for each service, wiring custom **RPC metrics** through the **Opentelemetry** Collector, and re-pointing **Kubernetes** auto-scaling to the Datadog metric source via **KEDA**.
- Hardened the encryption platform across **Go** and **Java SDK** stacks by shipping **Kafka** event-driven cache invalidation to eliminate new-customer encryption key creation latency. Replaced fail-fatal **LaunchDarkly** paths with a fault-tolerant client and stripped **S3** error fingerprinting to strip per-request identifiers from logs, decoupling a storage service from a feature-flag dependency and restoring downstream alert channels as appropriate incident triggers.
- Designed an event-driven solution for re-enabling **AWS** encryption keys marked for deletion by adding a new API endpoint to the **Java** backend of a **BYOK** service. Contributed to Frugal, a library built on **Apache Thrift**, by defining data types and service interfaces, enabling secure integration across polyglot **microservices** and preventing data loss from over **10,000** SEC reporting documents.
- Improved functionality of a tool that allows administrative-level users to perform queries against a service's internal data model. Implemented UI recognition of disabled actions for users by introducing a **Golang API** endpoint and changes to a **Dart** frontend, resulting in the completion of compliance requirements for many of Workiva's **6000+** clients, and an improvement in the support experience for **700+** engineers.

Amazon | Software Engineer Intern

May 2022 – Aug 2022

- Updated the **NodeJS back-end** and **React Typescript front-end** for a tool that automates and monitors quality assurance, to allow the user to create **1100%** more entries in an ML-based contact categorization engine.
- Intelligently populated a table on the front-end to allow for dynamic rendering of entries using **ReactJS**, utilizing documentation provided for an internal component library, resulting in a faster initial load time by up to **50%**.
- Utilized a pagination token in the data access layer to query from **DynamoDB** to support the increase in entries, and added encryption to all data sent through **HTTP** requests in an **AWS Lambda** function.

Education

University of Toronto | BSc. Computer Science

Sept 2019 – Dec 2023

- **Relevant coursework:** software design, operating systems, artificial intelligence, web development, algorithm design and complexity, database systems, computer graphics, computer security, computer networks

Projects

Interactive Animatronics Research Project | Python, Raspberry Pi



- Collaborated with a professor and team to develop a Python library leveraging LLMs for creating cost-effective, customizable animatronics kits, enhancing accessibility for educators and students through intuitive APIs.

HouseMate | MongoDB, Express, React, NodeJS



- Worked collaboratively to create a full-stack app designed to simplify cost splitting among a household.